Unit I - Introduction

Intelligence:

"The capacity to learn and solve problems."

Intelligence is the ability to learn, understand, reason, solve problems, and adapt to new situations in order to achieve goals.

Artificial Intelligence:

"AI is the branch of computer science which deals with Simulation of Human Intelligence in the machines."

- Computers with the ability to mimic or duplicate the functions of the human brain.
- computer systems able to perform tasks normally requiring human intelligence, such as visual perception, speech recognition, decision-making and translation between languages.

The history of AI

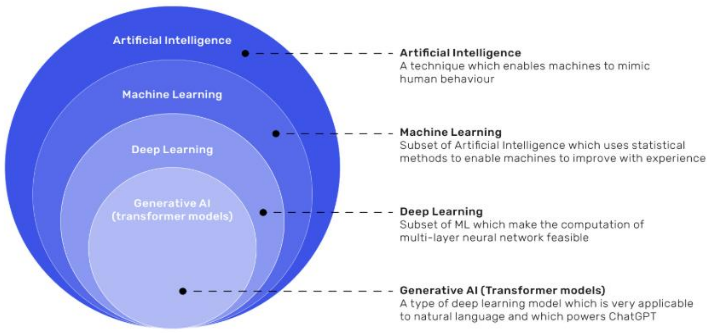
The idea of machines that can think for themselves isn't new. Concepts of artificial beings with intelligence stretch back decades, however, the modern field of AI truly began to take shape in the mid-20th century. Let's take a look at the history of AI as we know it:

- **The Seeds of AI (1940s-1950s)**: The invention of programmable computers in the 1940s sparked imaginations. **In 1950**, **Alan Turing proposed the "Turing Test,"** a way to gauge if a machine could exhibit intelligent behavior indistinguishable from a human. This was a critical philosophical and scientific step.
- **The Birth of a Field (1956)**: The Dartmouth Summer Research Project, organized by pioneers like **John McCarthy**, is widely considered the official birth of AI as an academic discipline. It was here that the term "**artificial intelligence**" was coined.
- **Early Successes and Challenges (1960s-1970s)**: Researchers developed early AI programs, like ELIZA, a chat bot that could simulate conversations, and Shakey the Robot, one of the first robots to reason about its environment. However, the complexity of creating true intelligence led to periods of reduced funding and progress, often called "AI Winters."
- **Revival and Growth (1980s-2000s)**: The development of expert systems and later, the rise of machine learning, breathed new life into AI research. Milestones like **IBM's Deep Blue defeating a chess** grandmaster in **1997** showcased AI's growing capabilities.
- **The Modern AI Boom (2010s-Present)**: Advances in computing power, the availability of massive datasets, and breakthroughs in deep learning, especially with neural networks, have fueled the current AI revolution. This era has seen the emergence of powerful tools that are transforming industries.

Subsets of Artificial Intelligence

AI can be classified into two main categories based on its capabilities and functionalities.

Figure 1. Subsets of Artificial Intelligence



Artificial Intelligence (AI):

Artificial Intelligence is the broad field that enables machines to mimic human intelligence and behavior. It includes problem-solving, decision-making, and understanding language. AI is the umbrella term covering all intelligent systems.

Machine Learning (ML):

Machine Learning is a subset of AI that allows machines to learn from data and improve performance without being explicitly programmed. It uses statistical techniques to identify patterns and make predictions based on experience.

Deep Learning (DL):

Deep Learning is a subset of Machine Learning that uses multi-layered neural networks to process large and complex data. It is especially effective in tasks like image recognition, speech processing, and natural language understanding.

Generative AI (Transformer Models):

Generative AI is a type of deep learning model that creates new content such as text, images, or code. Transformer-based models like ChatGPT excel at understanding and generating human language using large-scale data.

Based on Capability

Artificial Narrow Intelligence (ANI):

ANI is the only type of AI that exists today and is designed to perform a single specific task, such as image recognition, chatbots, or voice assistants. It does not have self-awareness or reasoning and works using data and algorithms. Poor data can cause bias or inaccurate results.

Artificial General Intelligence (AGI):

AGI is a future form of AI that is expected to think and learn like humans and perform multiple tasks. It would be adaptive, autonomous, and capable of improving from experience. AGI does not currently exist and raises ethical and safety concerns.

Artificial Superintelligence (ASI):

ASI is a theoretical AI that would surpass human intelligence in reasoning, creativity, and decision-making. It may be self-aware and operate beyond human control. Many experts warn it could pose serious risks to humanity if not properly regulated.

Based on Functionality

- **Reactive Machines:** Respond only to current inputs, no memory (e.g., Deep Blue).
- **Limited Memory:** Learn from past data (e.g., self-driving cars, chatbots).
- **Theory of Mind:** Understand emotions and social interactions (future).
- **Self-Aware AI:** Fully conscious AI (theoretical).

Weak AI and Strong AI

Weak AI (Narrow AI): This type of AI is designed to perform a specific task or a narrow set of tasks such as voice assistants or recommendation systems. It is good in one area like recommending products or recognizing speech but lacks general intelligence.

Weak AI: Built to **simulate** human intelligence.

Narrow Artificial Intelligence is limited to narrow (specific) areas like most of the AI we have around us today:

- Email spam Filters
- Text to Speech
- Speech Recognition
- Self Driving Cars
- E-Payment
- Google Maps
- Text Autocorrect
- Automated Translation
- Chatbots
- Social Media
- Face Detection
- Visual Perception
- Search Algorithms
- Robots
- Automated Investment
- NLP - Natural Language Processing
- Flying Drones
- IBM's Dr. Watson
- Apple's Siri
- Microsoft's Cortana
- Amazon's Alexa
- Netflix's Recommendations

Strong AI (General AI): It is a theoretical concept where AI can perform any intellectual task that a human can do. It shows human-like reasoning and understanding across multiple domains, making it capable of tackling a variety of tasks. Strong AI: Built to **copy** human intelligence.

- Strong AI
  Strong Artificial Intelligence is the type of AI that mimics human intelligence.

Strong AI indicates the ability to think, plan, learn, and communicate.
Strong AI is the theoretical next level of AI: **True Intelligence**.
Strong AI moves towards machines with self-awareness, consciousness, and objective thoughts.

- **Superintelligent AI**: It is a hypothetical form of AI that would surpass human intelligence in all areas. It would be capable of performing tasks more efficiently and effectively than humans.

Rationality refers to an AI system's ability to **make the best possible decision** based on the information it has.

A **rational agent**:

- Chooses actions that **maximize performance**
- Acts based on **available information**
- Aims for the **best possible outcome**

View	Description	Focus
Thinking Humanly	AI tries to replicate human thought processes	Understanding human mind
Thinking Rationally	AI aims to use logical reasoning and models	Logical correctness
Acting Humanly	AI behaves like humans (passes Turing Test)	Human-like behavior
Acting Rationally	AI acts to achieve the best outcome (rational agent)	Optimal decision-making

Cutting Edge Technologies:

Generative AI:
Generative AI creates new content such as text, images, music, or code instead of just analyzing data. It learns patterns from large datasets and generates original outputs based on user prompts. Examples include ChatGPT and DALL·E.

Large Language Models (LLMs):
LLMs are advanced AI models trained on massive amounts of text and code to understand and generate human language. They can perform tasks like answering questions, summarizing content, translating languages, and writing code. Many modern AI tools are powered by LLMs.

AI Agents:
AI agents are intelligent systems that can perceive their environment, make decisions, and take actions to achieve specific goals. They can plan tasks, reason through problems, interact with tools or systems, and adapt based on experience. Unlike basic chatbots, AI agents work more independently.

Agentic AI:
Agentic AI refers to AI systems that operate with a high level of autonomy, performing tasks without constant human input. These systems can manage complex workflows, interact with software tools and APIs, and make decisions on their own. Agentic AI is especially useful in software development for tasks like testing, refactoring code, and project management.

Applications of AI

1. Core Applications of AI

a) AI in Healthcare

- **Medical Imaging**: AI detects diseases like cancer or pneumonia from X-rays and MRI scans.
- **Virtual Health Assistants**: AI chatbots provide symptom checking and appointment scheduling.
- **Predictive Healthcare**: AI predicts patient risks such as heart disease or diabetes using health records.

Real-world examples we use:

- Google Health AI – Detects breast cancer from mammograms.
- Apple Watch & Fitbit – Uses AI to detect irregular heartbeats and activity patterns.
- Practo / 1mg (India) – AI-based symptom checkers and doctor recommendations.
- IBM Watson Health – Assists doctors in treatment planning**.**

b) AI in Education

AI personalizes learning and automates academic processes.

- Intelligent Tutoring Systems: AI adapts lessons based on student performance.
- Automated Evaluation: AI grades exams and assignments.
- Personalized Learning Platforms: Suggests courses and learning paths.

Real-world examples we use:

- Duolingo – AI personalizes language learning based on user mistakes.
- BYJU'S – AI adapts lessons to student learning speed.
- Google Classroom – AI helps teachers organize assignments and grading.
- ChatGPT – Used by students for explanations, summaries, and practice questions.

c) AI in Finance

AI enhances security, efficiency, and decision-making in financial systems.

- **Fraud Detection**: Identifies suspicious transactions in real time.
- **Algorithmic Trading**: AI predicts market trends and executes trades.
- **Credit Scoring**: AI evaluates loan eligibility using multiple data sources.

Real-world examples we use:

- **UPI & Banking Apps** – AI detects fraud in transactions.
- **PayPal** – AI prevents fraudulent payments.
- **Zerodha / Groww** – AI-based market insights and recommendations.
- **CRED** – Uses AI for credit analysis and personalized offers.

d) AI in Transportation

AI makes transportation safer and more efficient.

- **Self-Driving Vehicles**: Uses computer vision and sensors for navigation.
- **Traffic Management**: AI optimizes traffic signals to reduce congestion.
- **Route Optimization**: Suggests fastest delivery routes.

Real-world examples we use:

- **Google Maps** – AI predicts traffic and suggests fastest routes.
- **Uber / Ola** – AI calculates fares, matches drivers, and predicts demand.
- **Tesla Autopilot** – AI assists in self-driving features.
- **IRCTC** – AI predicts seat availability and waiting list chances.

2. New & Advanced AI Applications:

a) **Generative AI**

- Generative AI creates new content such as text, images, audio, and video.
- **Content Creation**: Generates articles, images, and marketing material.
- **Code Generation**: Assists developers by writing and debugging code.

- **Design & Media**: Creates music, animations, and 3D models.

Real-world examples we use:

- **ChatGPT** – Generates text, answers questions, and helps with coding.
- **DALL·E / Midjourney** – Creates images from text prompts.
- **Canva AI** – Generates designs, posters, and presentations.
- **GitHub Copilot** – Writes and suggests code for developers.

b) Agentic AI (Autonomous AI Agents)

Agentic AI can plan, decide, and act independently to achieve goals.

- **Task Automation**: AI agents manage emails, schedules, and reminders.
- **Multi-Agent Systems**: Multiple AI agents collaborate to solve complex problems.
- **Decision Making**: AI autonomously selects tools and executes tasks.

Real-world examples we use:

- **Auto-GPT** – AI agents that complete tasks autonomously.
- **Google Assistant / Alexa** – Books reminders, controls smart devices.
- **AI Customer Support Bots** – Resolve customer issues without human help.
- **ReAct Agents (Enterprise tools)** – Automatically handle workflows like ticket resolution.

c) AI in Workplace & Productivity

AI improves efficiency and reduces manual work.

- **Document Automation**: AI summarizes reports and meeting notes.
- **Smart Assistants**: Helps with writing, coding, and data analysis.
- **HR Automation**: AI screens resumes and predicts employee attrition.

Real-world examples we use:

- **Microsoft Copilot** – Helps in Word, Excel, Outlook, and Teams.
- **Notion AI** – Writes summaries, meeting notes, and project plans.
- **Grammarly** – AI improves writing and grammar.
- **Zoom AI Companion** – Summarizes meetings and action points.

d) AI in Smart Cities

AI helps manage urban infrastructure and public services.

- **Smart Surveillance**: AI monitors public safety.
- **Energy Management**: Optimizes power usage and reduces wastage.
- **Waste Management**: AI predicts garbage collection needs.

Real-world examples we use:

- **Smart Traffic Signals (India & Singapore)** – AI adjusts signals based on traffic flow.
- **CCTV with AI Analytics** – Used for crowd monitoring and crime detection.
- **Smart Parking Systems** – AI shows available parking spots.
- **Energy Smart Meters** – AI optimizes electricity usage.

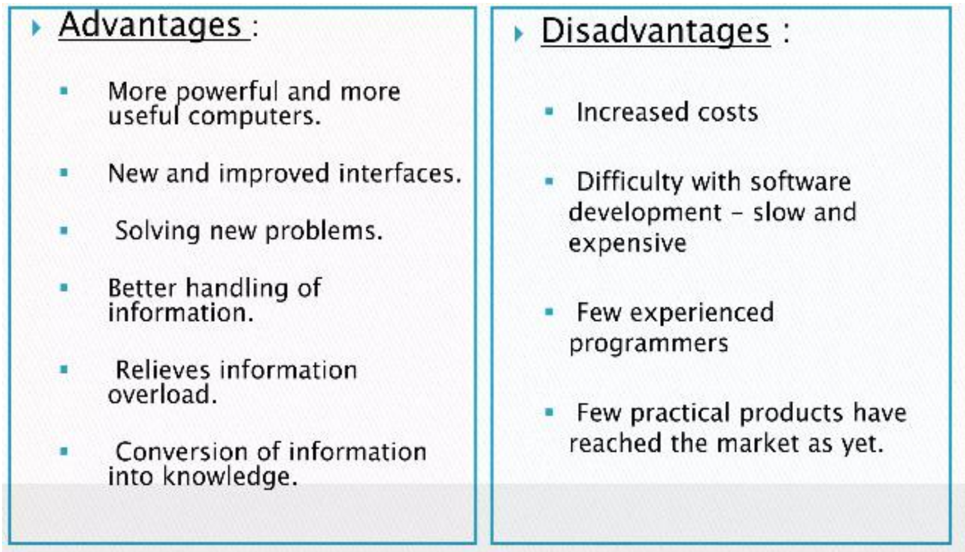
e) AI in Science & Research

AI accelerates discovery and innovation.

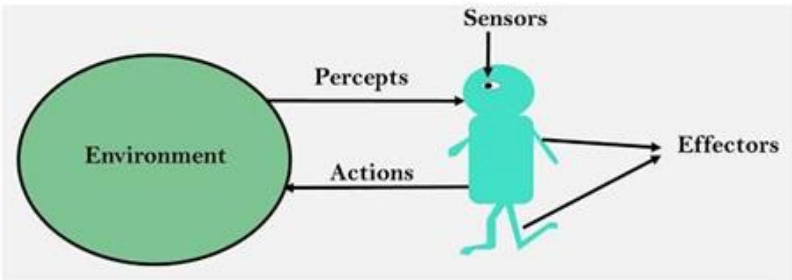
- **Drug Discovery**: AI identifies potential medicines faster.
- **Climate Research**: Predicts climate patterns and natural disasters.
- **Scientific Simulations**: AI models complex systems like galaxies or molecules.

Real-world examples we use:

- **AlphaFold (DeepMind)** – Predicts protein structures.
- **NASA AI Systems** – Analyzes space data and detects anomalies.
- **AI Drug Discovery Platforms** – Used by pharma companies to develop medicines.
- **Climate AI Models** – Predict floods, cyclones, and climate change patterns.



Intelligent Agents



An agent is anything that can be viewed as :

-perceiving its environment through sensors and

-acting upon that environment through actuators

Note : Every agent can perceive its own actions (but not always the effects)

OR

"An Agent refers to an entity that perceives its environment through sensors and acts upon that environment using actuators."

Agents in Artificial Intelligence

• Artificial intelligence is defined as study of rational agents. A rational agent could be anything which makes decisions, like a person, firm, machine, or software. It carries out an action with the best outcome after considering past and current percepts (agent's perceptual inputs at a given instance). An AI system is composed of an agent and its environment.

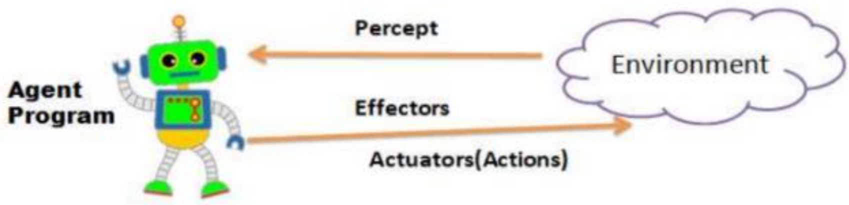
• The agents act in their environment. The environment may contain other agents.

• We use the term percept to refer to the agent's perceptual inputs at any given instant. An agent's percept sequence is the complete history of everything the agent has ever perceived

In general, an agent's choice of action at any given instant can depend on the entire percept sequence observed to date, but not on anything it hasn't perceived Operations which is performed by the agent is Sense from the environment or atmosphere with the help of different sensors.

Structure of an agent

Agent→Percept→Decision→Action→Environment



Percept→Current state + History

To understand the structure of Intelligent Agents, we should be familiar with Architecture and Agent Program.

• Architecture is the machinery that the agent executes on. It is a device with sensors and actuators, for example : a robotic car, a camera, a PC.

• Agent program is an implementation of an agent function. An agent function is a map from the percept sequence(history of all that an agent has perceived till date) to an action.

Agent = Architecture + Agent Program

Things that machine/agent have sensed in present, if it looks similar to something in past or not, it takes some decisions on the basis of it.

Architecture: The machinery that an agent execute on.

Agent Program: An implementation of an agent function.

Various types of Agent:

- Human agent:
  A human agent has sensory organs like eyes, ears, nose, toungue, and skin parellel to other sensors and hands, legs, vocal tract, and so on for actuators.
- Robotic agent:
  A robotic agent might have cameras and infrared range finders for sensors and various motors for actuators.
- Software Agent:
  A software agent has Keystrokes, file contents, received network packages which act as sensors and displays on the screen, files, sent network packets acting as actuators.

Agent terminology:

1. **Performance measure of agent:** It is the criteria determining the success of an agent.

2. **Behaviour/action of agent:** It is the action performed by an agent after any specified sequence of the precepts.

3. **Percept:** It is defined as an agent's perceptual inputs at a specified instance.

4. **Percept sequence:** It is defined as the history of everything that an agent has perceived till date.

5. **Agent function:** It is defined as a map from the precept sequence to an action.

Agent function, a = F(p), where p is the current percept, a is the action carried out, and F is the agent function.

where p is the current percept, a is the action carried out, and F is the agent function. F maps precepts to actions
F :P → A

where P is the set of all precepts, and A is the set of all actions. Generally, an action may be dependent of all the precepts observed, not only the current percept,

ak = F(p0 p1 p2 ... pk) Where

1. p0 , p1, p2, ... ,pk is the sequence of percepts recorded till date.

2. ak is the resulting action carried out and F now maps percept sequences to action

F :P * → A

Environment and its Properties

Environment: The external context or world in which an agent operates. It includes everything that an agent can perceive, interact with, or affect.

Types of Environment

1. Fully Observable vs Partially Observable

2. Competitive vs Collaborative

3. Single-agent vs Multi-agent

4. Static vs Dynamic

5. Discrete vs Continuous

6. Episodic vs Sequential

1. **Fully Observable vs Partially Observable.**

When an agent sensor is capable to sense or access the complete state of an agent at each point of time, it is said to be fully observable environment else it is partially observable.

"Environment which an agent has complete access to all relevant, necessary information through the sensors for decision-making at every point in time are known as fully observable environments"

Maintaining fully observable environment is easy as there is no need to keep track of the history of the surroundings.

Partially Observable

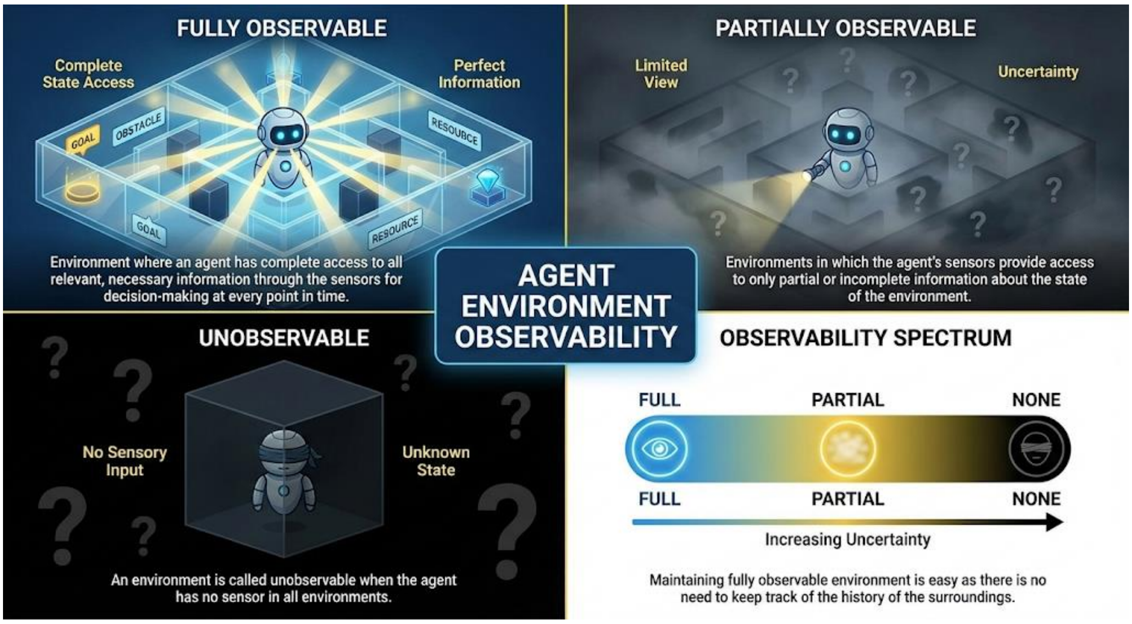
"Environments in which the agent's sensors provide access to only partial or incomplete information about the state of the environment at each point in time. are known as partially observable environments."

An environment is called unobservable when the agent has no sensor in all environments.

Examples:

Chess: The board is fully observable and so are the opponent's moves.

Driving: The environment is partially observable because what's around the corner is unknown.



2. Deterministic vs Stochastic

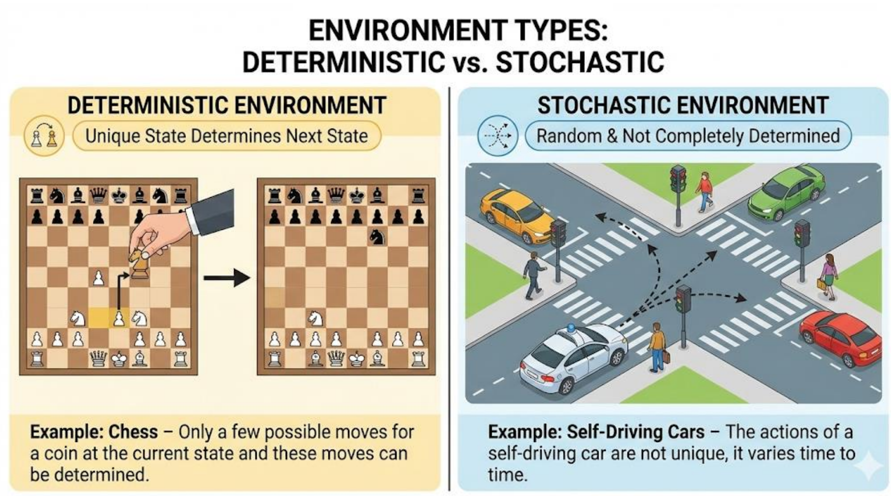
When a uniqueness in the agent's current state completely determines the next state of an agent, the environment is said to be **deterministic.**

Stochastic: The stochastic environment is random in nature which is not unique and cannot be completely determined by the agent.

Examples:

Chess – there would be only a few possible moves for a coin at the current state and these moves can be determined.

Self-Driving Cars- the actions of a self-driving car are not unique it varies time to time.



3.Competitive vs Collaborative

An agent is said to be in a **competitive** environment when it competes against another agent to optimize the output. The game of chess is competitive as the agents compete with each other to win the game which is the output.

Collaborative :

An agent is said to be in a collaborative environment when multiple agents cooperate to produce the desired output.

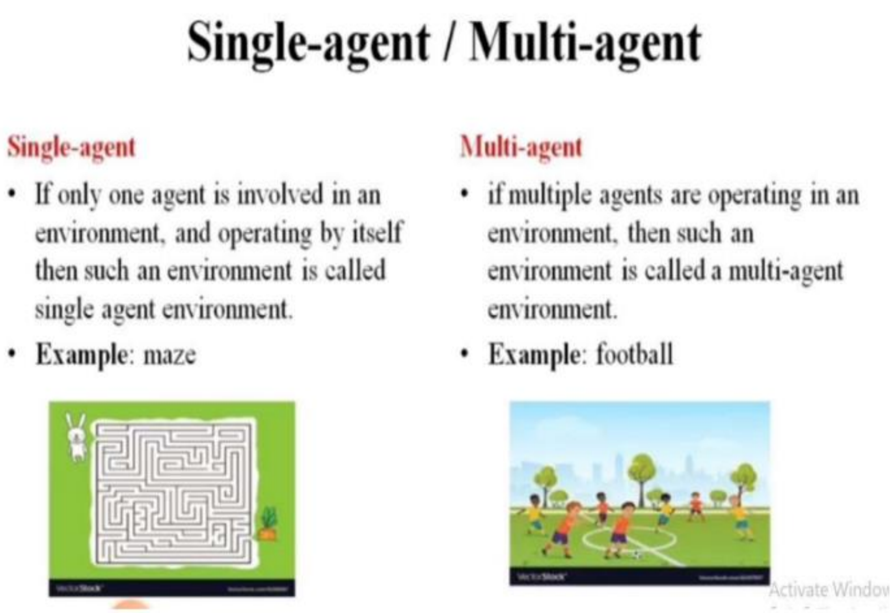
When multiple self-driving cars are found on the roads, they cooperate with each other to avoid collisions and reach their destination which is the output desired.

4. Single-agent vs Multi-agent

An environment consisting of only one agent is said to be a single- agent environment.

- **Single-agent**: A person left alone in a maze is an example of the single-agent system.
- **Multi-agent**: An environment involving more than one agent is a multi-agent environment.

The game of football is multi-agent as it involves 11 players in each team.



5.Dynamic vs Static

Dynamic: An environment that keeps constantly changing itself when the agent is up with some action is said to be dynamic.

A roller coaster ride is dynamic as it is set in motion and the environment keeps changing every instant.

Static: An idle environment with no change in its state is called a static environment. An empty house is static as there’s no change in the surroundings when an agent enters.

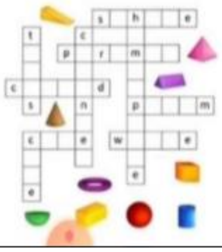
Static / Dynamic

Static

- The environment does not change while an agent is acting
- **Example**: Crossword puzzles

Dynamic

- The environment may change over time
- **Example**: roller coaster ride
  Taxi driving



6.Discrete vs Continuous

Discrete: If an environment consists of a finite number of actions that can be deliberated in the environment to obtain the output, it is said to be a discrete environment.

Ex: The game of chess is discrete as it has only a finite number of moves. The number of moves might vary with every game, but still, it’s finite

Continuous: The environment in which the actions are performed cannot be numbered i.e. is not discrete, is said to be continuous.

Ex: Self-driving cars are an example of continuous environments as their actions are driving, parking, etc. which cannot be numbered.

Discrete / Continuous

Discrete

- The environment consists of a finite number of actions that can be deliberated in the environment to obtain the output.
- **Example**: chess

Continuous

- The environment in which the actions performed cannot be numbered ie.,is not discrete, is said to be continuous.
- **Example**: Self-driving cars

TYPES OF AGENTS:

1. Simple Reflex Agent
2. Model Based Agent
3. Goal Based Agent
4. Utility Based Agent

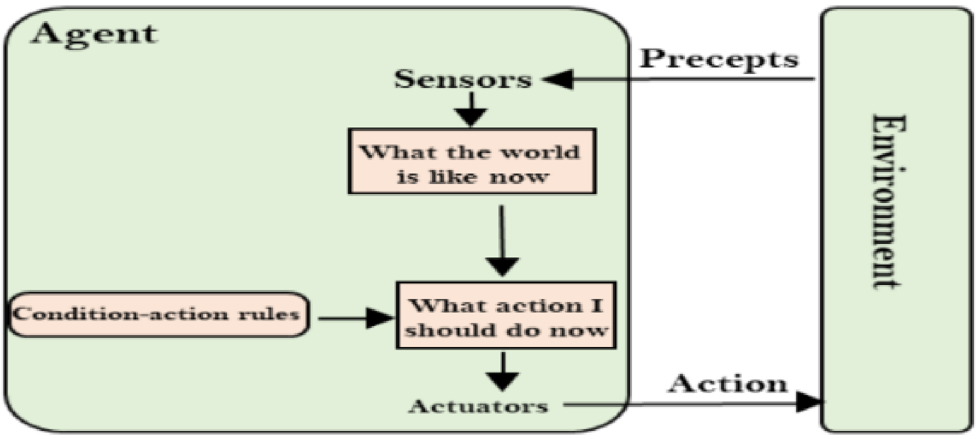
1. SIMPLE REFLEX AGENT

1. The Simple reflex agents are the simplest agents. These agents take decisions on the basis of the current precepts and ignore the rest of the percept history.

2. These agents only succeed in the fully observable environment.

3. The Simple reflex agent does not consider any part of precepts history during their decision and action process.

4. The Simple reflex agent works on Condition-action rule, which means it maps the current state to action. Such as a Room Cleaner agent, it works only if there is dirt in the room.



Problems for the simple reflex agent design approach:

1. They have very limited intelligence
2. They do not have knowledge of non-perceptual parts of the current state
3. Mostly too big to generate and to store.
4. Not adaptive to changes in the environment.

2. Model-based reflex agent

The Model-based agent can work in a partially observable environment, and track the situation. A model-based agent has two important factors:

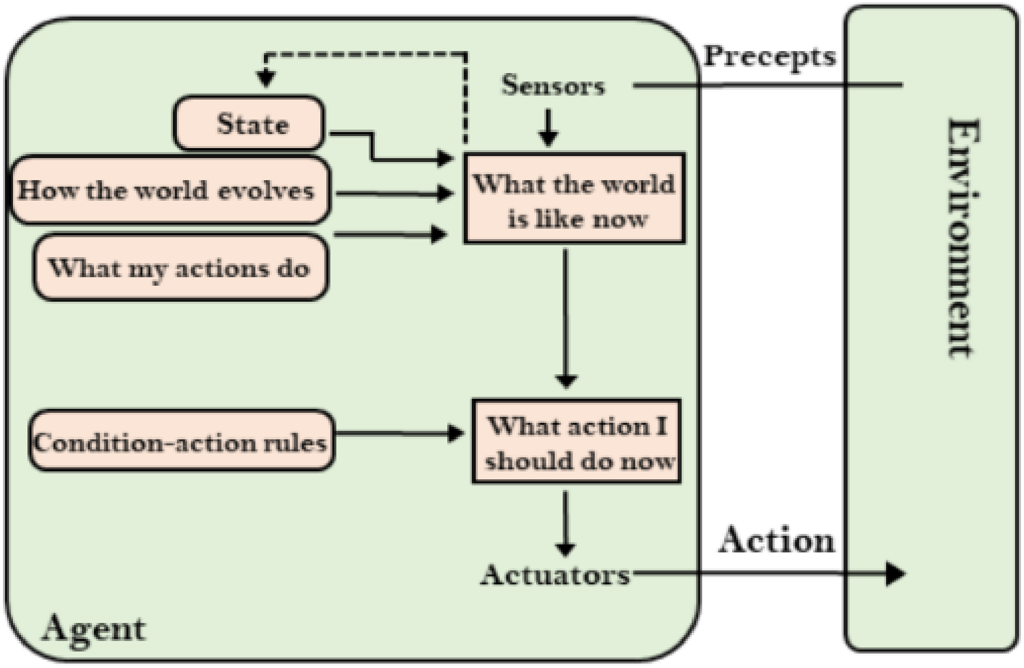
- Model: It is knowledge about "how things happen in the world," so it is called a Model-based agent.
- Internal State: It is a representation of the current state based on percept history.

These agents have the model, "which is knowledge of the world" and based on the model they perform actions.

Updating the agent state requires information about:

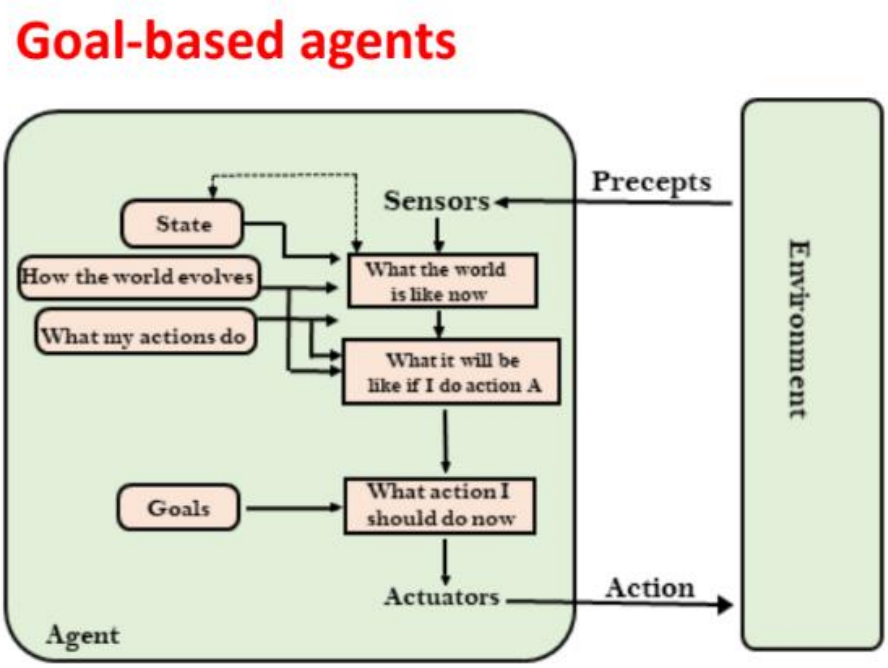
- How the world evolves
- How the agent's action affects the world.

The Model-based reflex agent



3. Goal-based agents

- The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
- The agent needs to know its goal which describes desirable situations.
- Goal-based agents expand the capabilities of the model-based agent by having the "goal" information.
- They choose an action, so that they can achieve the goal.
- These agents may have to consider a long sequence of possible actions before deciding whether the goal is
- achieved or not. Such considerations of different scenario are called searching and planning, which makes an agent proactive.



Goal based agent is based on two techniques:

1. Searching

2. Planning

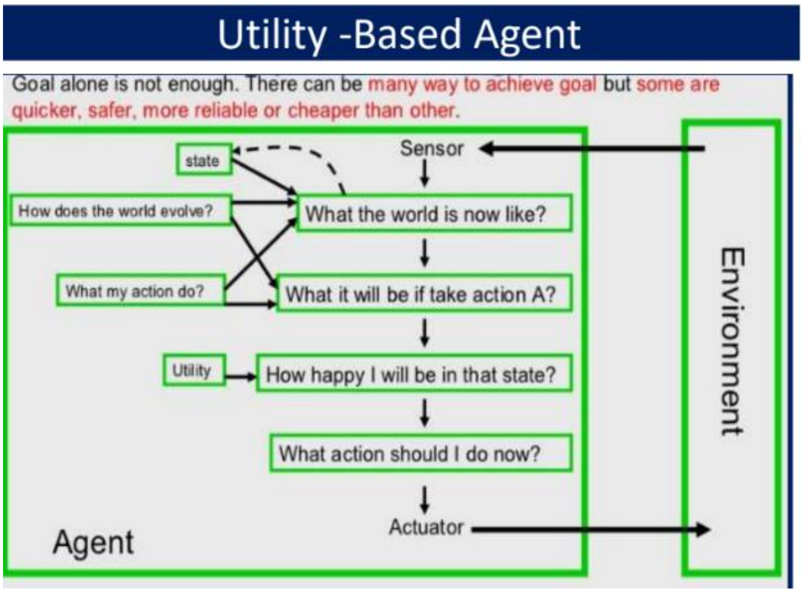
Searching: Finding the route (Shortest path)

Planning: To reach the goal state the agent has to plan.

Example: Planning for a trip

4. Utility -Based Agent

- Utility-based agents are a type of intelligent agent in artificial intelligence (AI) that make decisions based on a utility function.
- Utility function: This function measures the degree of satisfaction or utility associated with different possible
  outcomes.
  Unlike simpler agents, which might react to stimuli or follow predefined goals, utility-based agents evaluate multiple potential actions and select the one that maximizes their overall utility.
- Components of Utility-Based Agents



PEAS Representation of an agent:

Goals of agent

- High performance
- Optimized result
- Rational action

Factors for agents(PEAS):

P: Performance

E: Environment

A: Action

S: Sensors

P: Performance

It shows how the systems achievements is measured.

Ex: Automatic cars-They are autonomous which takes decisions on their own.

- Safety-reach the destination safely.
- Comfortness.
- Proper sensing of the obstacles

E: Environment

It represents what the agent is iteracting with.

- Automatic car is moving in the environment.
- People moving on road.
- Cars moving in the same or opposite direction.
- Animals moving on the road.

A: Action/Actuators:

It is what produces the outputs of the system.

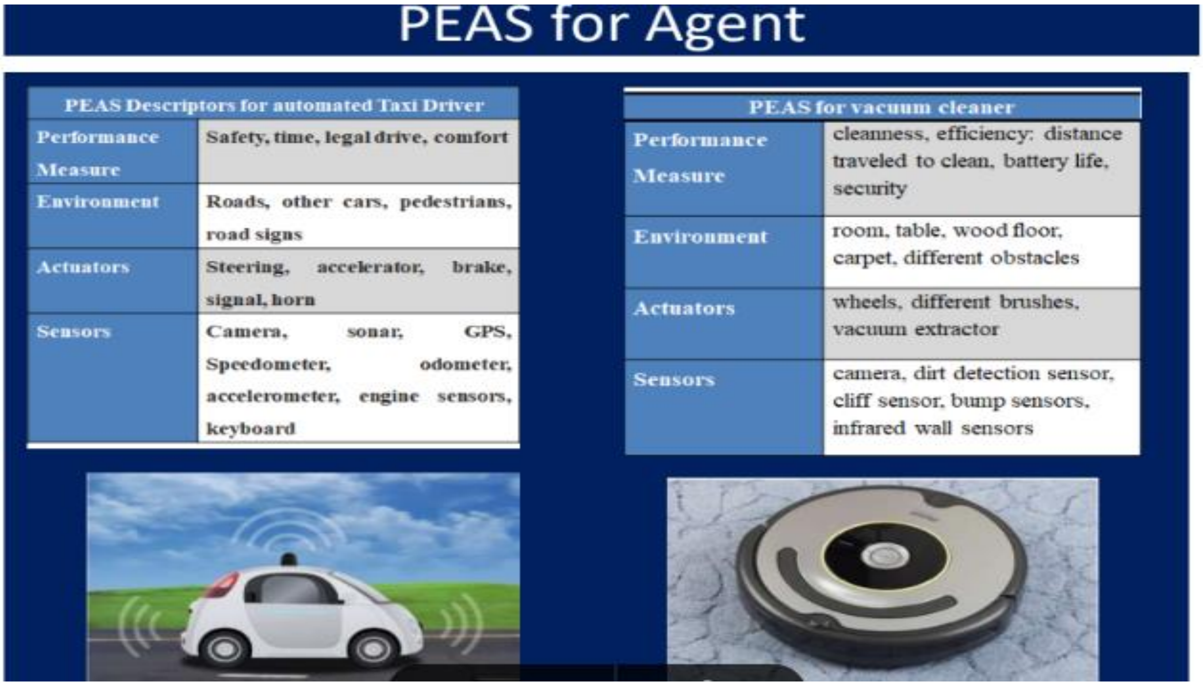
- Moving the steering-left or right
- Using brakes, accelerators and horn when obstacles are sensed

S: Sensors:

It provides the inputs to the system

Example: Camera, sonar, GPS, Speedometer, odometer, accelerometer, engine sensor ,keyboard, dirt detection sensor.

- Wipers should wipe the glass when there is rain



Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical Diagnosis System	Accurate diagnosis, healthy patient, reduced cost	Patient, hospital, medical staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard input of symptoms, test results, patient answers
Satellite Image Analysis System	Correct image classification	Satellite images, ground stations	Display of categorized images	Color pixel arrays
Part-Picking Robot	Correct placement of parts, efficiency	Conveyor belt, bins, factory floor	Robotic arm, gripper	Camera, joint angle sensors
Refinery Controller	High purity, maximum yield, safety	Refinery plant, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English Tutor	Improved student test scores	Students, testing agency	Display of exercises, hints, corrections	Keyboard input, microphone

Problem Solving

In Artificial Intelligence (AI), the term problem solving is given to the analysis of how computers can be made to find solutions in well-circumscribed domains. Since puzzles and games have explicit rules, these are often used to explore ideas in AI (whether the insights thus gained will transfer to less formal domains is another issue). A good example of a problem- solving domain is the “Tower of Hanoi” puzzle

Problem solving is defined as the way in which an agent finds a sequence of actions that achieves its goals, when no single action will do. Problem formulation usually requires abstracting away real-world details to define a state space that can feasibly be explored. This step of abstraction is performed by an agent called as a problem-solving agent.

STATES:

1. **Initial State**: States where a given episode of problem solving starts.

2. **Goal or Solution State:** States that are considered solution to the problems.

3. Failure or impossible states: In some domains, there are states with the property that if they are ever encountered, the problem solving is considered as failure.

Structure of problem solving agent

1. **Goal state:** A state that describes the objective that the agent is trying to achieve is called as the goal state of the agent.
2. **Action:** When there is a transition between the world states an action is said to be performed

Steps in problem solving by problem solving agent

1. **Goal formulation:** On the basis of the current situation and agent's performance measure, it is the first step in problem solving. Agent's task is to find out sequence of actions that will get it to the goal state. Goals help organize behavior by limiting the objectives the agent is trying to achieve.

2. **Problem formulation:** It is a process of deciding what actions and states to consider given a goal.

3. **Choosing the best sequence:** An agent with several immediate options of unknown value can decide what to do by first examining different possible sequences of actions that lead to the states of known value and then choosing the best sequence.

Problem Formulation

Problem formulation is the process of deciding what actions and states to consider given a goal.

ALGORITHM

Step 1: Analyse the problem to get the starting state and goal state.

Step 2: Find out the data about the starting state, goal state.

Step 3: Find out the production rules from initial database for progressing the problem to goal state.

Step 4: Select some rules from the set of rules that can be applied to data.

Step 5: Apply those rules to the initial state and proceed to get the next state.

Step 6: Determine some new generated states after applying the rules. Accordingly, make them as current state.

Step 7: Finally, achieve some information about the goal state from the recently used current state and get the goal state.

Step 8: Exit.

Problem formulation of the Eight Tile Puzzle

1. Problem Definition

• The 8-Tile Puzzle consists of a 3x3 grid containing 8 numbered tiles (1 to 8) and one empty space. The goal is to rearrange the tiles from an initial scrambled state into a specified goal state by sliding tiles into the empty space.

2. State Representation

• Each configuration of the grid is a state. The empty space determines the possible moves.

• A state is represented as a 3x3 matrix (or a 1D array with indices).

• The successor states are generated by moving a tile into the blank space.

States: It specifies the location of each of the eight tiles and the blank in one of the nice squares.

• Initial state: Any state can be designated as the initial state.

• Goal: Many goal configurations are possible.

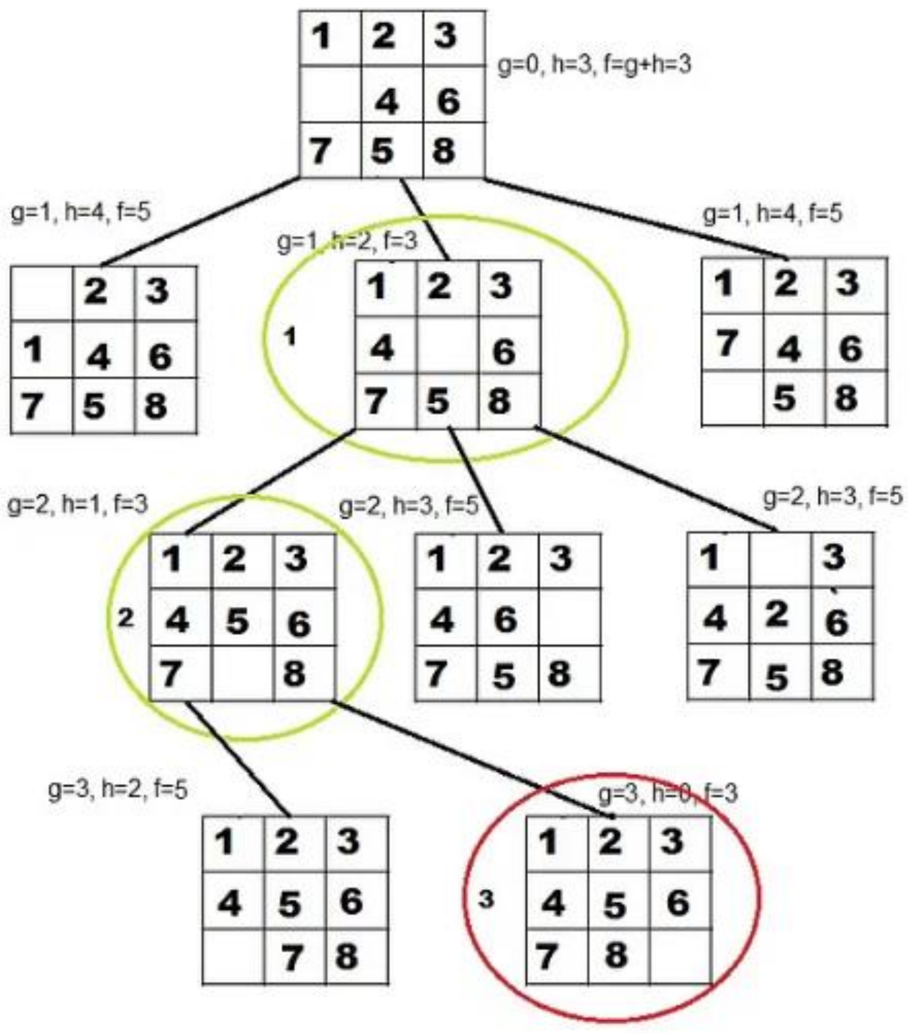
• Legal moves (or state): They generate legal states that result from trying the four actions:

- Blank moves left
- Blank moves right
- Blank moves up
- Blank moves down

• Path cost: Each step costs 1, so the path cost is the number of steps in the path.

• Applications of the 8-Tile Puzzle:

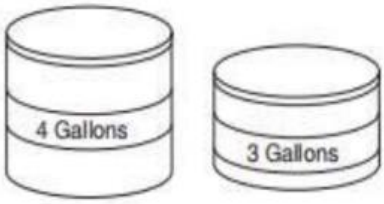
1. Used to test search algorithms in AI.
2. Helps in understanding state-space representation.
3. Extends to 15-Puzzle, 24-Puzzle, and N-Puzzle problems.



Problem formulation of the Water Jug Problem

Problem definition: You have given two jugs one with capacity of four gallons and other with capacity of three gallons none is having any measuring marks on it. There is the pump that can be used to fill water in jug.

Problem formulation Steps:



- 1. Describe the states
- 2. Identify the initial state
- 3. Identify the goal state
- 4. Identify set of rules (all possible actions)
- 5. Find the solution path in the state space

Solution:

Now, we need to solve the problem such that there is two gallons of water in four gallons of jug.

We will have to keep track of the amount of water in each of the jugs after we perform a particular action. Hence, we should represent the state such that it reflects the amount of water in each of the individual jugs.

• Representation of state: For the mentioned problem,

1. The state can be described as ordered pair of integer (x, y) such that 'x' is the amount of water in four gallon of jug, hence 'x' can take values 0, 1, 2, 3, 4 gallons.

Note: Since 'x' is the amount of water in four gallon of jug, the jug can be empty, i.e. has 0 gallons of water or can have 1/2/3/4 gallons of water, respectively). 'y' is the amount of water in three

gallons of jug. Similarly can take values as 0, 1, 2, 3 ...

- Initial state: Initially, we assume that both the jugs are empty. That is both the jugs contain zero gallons of water. This initial situation (also called as initial state) can be represented using state representation notation as follows:
  SI = (0, 0) Note that initial state is represented by SI.
- Goal state: We need two gallons of water in four gallons of jug as the goal but there can be any amount of water in three gallons of jug. Hence goal state is the set of states.
  SG = {(2, 0), (2, 1), (2, 2), (2, 3)}
- This notation indicates that there must be two gallons of water in four gallons of jug but there can be 0/1/2/3 gallons of water in three gallons of jug which we are not concerned. Hence, goal state represented by SG is set of states as shown. Hence, as the goal the agent can achieve any one of the states that belongs to this set.

Set of rules: Now, we write set of rules that will be applied to current state so as to change the current state and go to the next state. To write these set of rules one has to simply think of all possible combinations that can occur,

Rule 1: Fill in 4G Jug completely

$(x, y) \rightarrow (4, y)$ if $(x < 4)$

Rule 2: Fill in 3G Jug completely

$(x, y) \rightarrow (x, 3)$ if $(y < 3)$

Rule 3: Pour some water from 4G Jug to ground

$(x, y) \rightarrow (x-d, y)$ if $(x > 0)$

Rule 4: Pour some water from 3G jug to ground

$(x, y) \rightarrow (x, y-d)$ if $(y > 0)$

Rule 5: Empty 4G jug

$(x, y) \rightarrow (0, y)$ if $(x > 0)$

Rule 6: Empty 3G jug

$(x, y) \rightarrow (x, 0)$ if $(y > 0)$

Rule 7: Pour some water from 3G jug to 4G jug until 4G jug is FULL

$(x, y) \rightarrow (4, y - (4 - x))$ if $(y > 0)$ and $(x + y) \geq 4$

Rule 8: Pour some water from 4G jug to 3G jug until 3G jug is FULL

$(x, y) \rightarrow (x - (3 - y), 3)$ if $(x > 0)$ and $(x + y) \geq 3$

Rule 9: Pour all water from 3G jug to 4G jug

$(x, y) \rightarrow (x + y, 0)$ if $(y > 0)$ and $(x + y) \leq 4$

Rule 10: Pour all water from 4G jug to 3G jug

$(x, y) \rightarrow (0, x + y)$ if $(x > 0)$ and $(x + y) \leq 3$

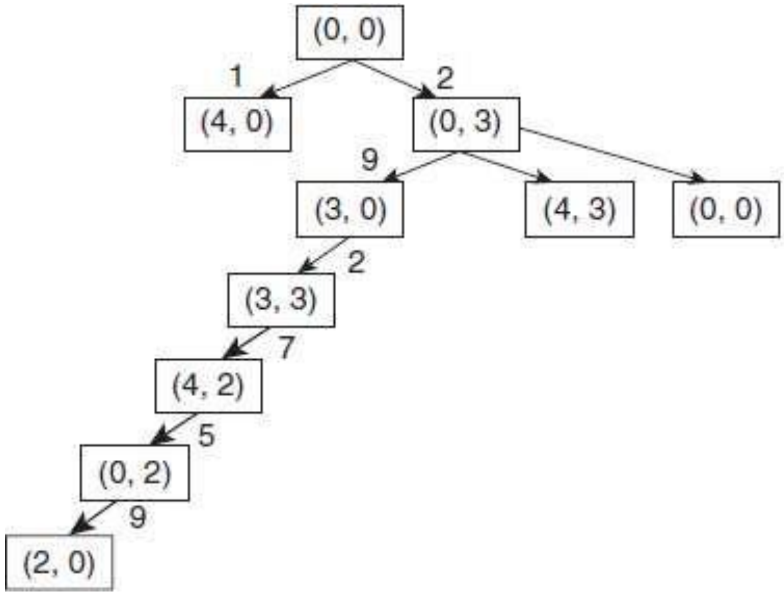
we will apply these set of rules to the initial state to generate the goal state. These set of rules will be applied to each state until we reach the goal. Hence, we

• Generate a tree which represents all the possible states in the problem, and this is called as state space.

• In a state space, we will find a path to the goal this is called a solution path. Solution path is found from state space by state space search.

• This approach of solution is called as state-space approach.

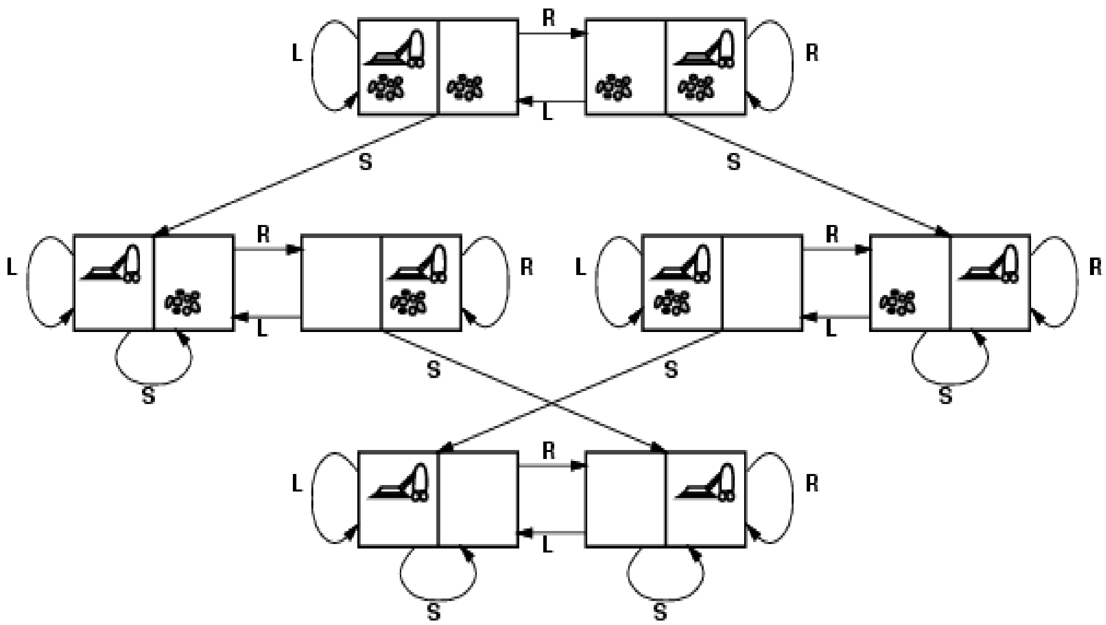
Solution Tree: To represent all the possible states in the state space.



Vacuum Cleaner Problem

The Vacuum Cleaner Problem is a classic AI problem used to demonstrate simple agent-based decision-making.

Below is a step-by-step explanation based on the given diagram:



Step 1: Understanding the Components

• The diagram represents different states of a vacuum cleaner in a two- room environment.

• Each box represents a possible state of the vacuum cleaner.

• The vacuum cleaner can be in one of two locations (Left or Right).

• The rooms can either be Clean or Dirty.

Actions available to the vacuum cleaner:

Suck (S): Cleans the room.

Move Left (L): Moves the vacuum to the left room.

Move Right (R): Moves the vacuum to the right room.

Step 2: Identifying the Initial State

•The vacuum cleaner starts in a certain location.

•The rooms might be clean or dirty.

Step 3: Agent Perception and Action

• The vacuum perceives whether the current room is dirty or clean.

• If the room is dirty, it performs the Suck (S) action to clean it.

• If the room is clean, it decides whether to move left (L) or right(R).

Step 4: State Transitions

• If the vacuum sucks the dirt, the state changes to a clean room.

• If the vacuum moves left or right, it transitions to a new state.

Step 5: Goal of the Vacuum Cleaner

• The goal is to clean both rooms efficiently.

• The agent follows a decision-making process to reach a state where both rooms are clean.

Step 6: Problem Representation as a Search Tree

• The diagram shows different possible states and transitions of the vacuum cleaner.

• Each node represents a state (position of the vacuum and cleanliness of the rooms).

• Each arrow represents an action leading to a new state.

Step 7: Optimal Solution

• The optimal solution minimizes the number of moves and suck actions needed to clean both rooms.

• The vacuum cleaner should avoid unnecessary movements once both rooms are clean.